EDUSPARK EXAMS CONSULT

SCIENCE
Essay and Objective
2hour 10minutes

1&2

Name: _____

Index number: _____

Date: _____

B.E.C.E 2027 STEPPING STONE MOCK – SEPTEMBER, 2026

SCEINCE [100 marks]

*Write your **name** and **date** in **ink** in the spaces provided above*
DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

- ✓ *This booklet consists of two papers, Paper **2** and paper **1***
- ✓ *Answer paper **2** in your answer booklet and paper **1** in the objective test answer sheet*
- ✓ *Paper **2** will last 1hour 25mins.*
- ✓ *Paper **1** will last* 45minutes.
- ✓ ***Do not start paper 1 until you are told to do so***

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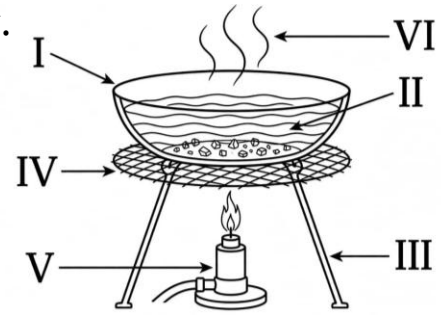
PAPER 2 - ESSAY [100 marks]

This paper is made up two sections; **Section A** and **B**. Answer Question 1 in section **A** and any other **three** questions from Section **B**.

SECTION A [40 marks]

Answer **all** questions from this section.

1. a) In the Integrated Science laboratory, Kofi set up the apparatus shown in the figure below to separate a certain mixture he was given. Study the diagram carefully and use it to answer the questions that follow.



(i) Name the parts labelled I, II, III, IV, V and VI.

(ii) What method of separation is Kofi using?

(iii) What will Kofi obtain in part I after heating the mixture in II for some time?

(iv) State the function of the parts labelled:

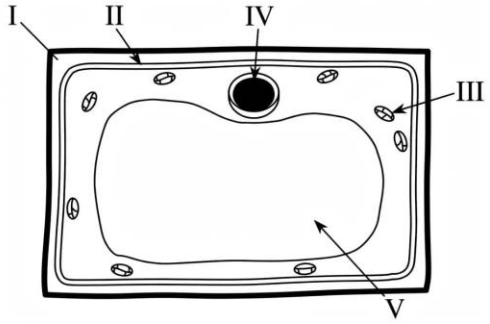
(α) IV

(β) V

(v) Why can the component obtained in I not be obtained by filtration of the mixture in II?

(vi) What change would Kofi need to make to the apparatus if he also wanted to collect the component that escapes at VI? What would the new method be called? **[10marks]**

b) The figure below is an illustration of a plant cell. Study the figure carefully and answer the questions that follow.



(i) Name the parts labelled I, II, III, IV and V. [5 marks]

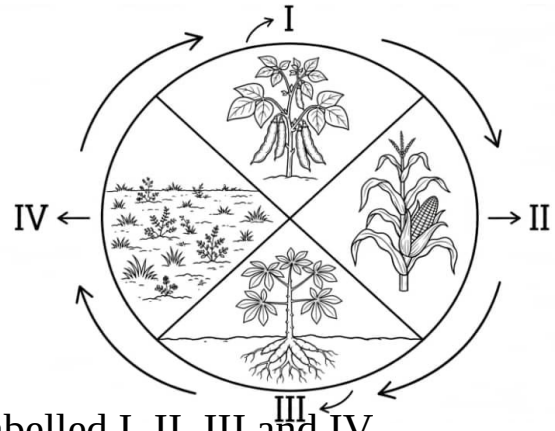
(ii) Give **one** function each of the parts labelled III, IV and V. [3 marks]

(iii) State:

(α) one part that is present in the plant cell but absent in an animal cell;

(β) one difference between the vacuole in a plant cell and that in an animal cell. **[10 marks]**

c) The figure below is an illustration of a crop rotation system. Study the figure carefully and answer the questions that follow.



(i) Name the crops/plots labelled I, II, III and IV.

(ii) To which group of crops does the crop labelled

(α) I belong?

(β) II belong?

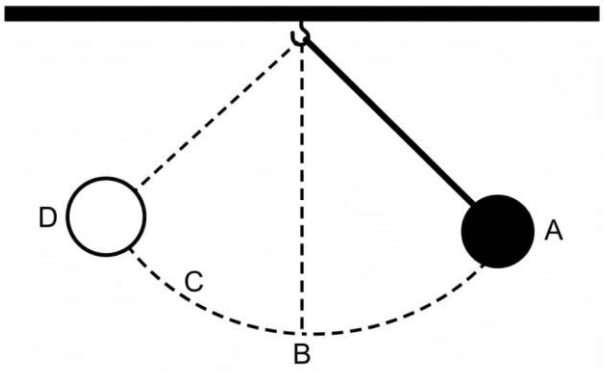
(γ) III belong?

iii) A farmer decides to plant the crop labelled II on the same plot for 4 consecutive years without rotating. Predict two problems he is likely to face.

iv) Which of the labelled crops helps to improve soil fertility naturally? Explain how it does it.

v) Why is the plot labelled IV important in this rotation system even though no crop is harvested from it? **[10marks]**

d) Study the diagram below and use it to answer the questions that follow.



i) Identify the diagram above.

ii) Identify the conversion of energy involved in the diagram, from A to D.

iii) Which type of energy is greatest at position B

iv) Where in the diagram is potential energy equal to zero?

v) What type of energy is the object having at point D?

vi) If the mass of the object is 2.5kg and its raised 10m above the ground at point **D**, what is the magnitude of the energy possessed by the object? [Take g = 10ms^2] **[10marks]**

SECTION B [60 marks]

Answer **three** questions **only** from this section.

2. (a) Give ***two*** examples of each of the following types of substances:

(i) element;

(ii) compound;

(iii) mixture. [6 marks]

(b) i. Explain the following

α) renewable energy source [1mark]

β) non-renewable energy source [1mark]

ii. List ***two*** examples each of renewable energy source and non-renewable energy source [2mark]

(c) (i) What is a cell? [1mark]

(ii) Draw and label a typical animal cell to show: nucleus, cell membrane and cytoplasm. [2marks]

(iii) State ***one*** function of the nucleus. [2marks]

(d) i) Explain the following terms associated with animal feeding.

α) Ration

β) Supplement [3marks]

ii) List four most common ways of feeding farm animals. [2marks]

3. (a) Name ***two***:

(i) deficiency diseases in humans;

(ii) diseases caused by fungi in humans. [4 marks]

(b) (i) What is a binary compound? [2marks]

(ii) Write the chemical formula of each of the following compounds:

(α) Sodium chloride;

(β) Ammonia;

(δ) Hydrochloric acid

(γ) Carbon dioxide. [4marks]

(c) (i) What is a balanced ration in poultry feed? [2marks]

(ii) Give ***three*** effects of malnutrition in poultry birds. [3 marks]

(b) i) Name the planets that constitute the inner planets of the solar system. [2marks]

ii) Mention three properties of the inner planets. [3marks]

4. (a) Name ***two***:

(i) micro-nutrients of plants;

(ii) macro-nutrients of plants. [4 marks]

(b) i) Briefly describe a self-sustaining ecosystem and give two examples. [3marks]

ii) Distinguish between biotic and abiotic component of an ecosystem [2marks]

(c) (i) Distinguish between a colloid and a suspension. [2marks]

(ii) Classify the following as solution, colloid or suspension:

(α) Salt and water;

(β) Milk and water;

(γ) Sand and water. [3 marks]

(d) State ***two*** effects of poor waste management on human health. [2 marks]

(e) (i) What is work input? [2marks]

(ii) Explain how work output affects the efficiency of a machine if work input is fixed. [2marks]

5. (a) (i) What is green economy? [2marks]

(ii) State ***three*** climate change adaptation measures that you can apply to conditions in your community. [3marks]

(b) (i) Name the main stages involved in electricity power supply. [3marks]

(ii) What is the function of a capacitor in an electronic circuit? [2marks]

(c) (i) State ***three*** non- metals in the first twenty elements. [3 marks]

(ii) Give ***three*** properties of non-metal. [3 marks]

(d) (i) Mention ***two*** breeding habit of a house fly. [2 marks]

(ii) State ***two*** possible diseases that housefly can cause. [2 marks]

PAPER 1

OBJECTIVE TEST *45 Minutes*

*Answer **All** Questions*

Each question is followed by four options lettered A to D. Find out the correct option for each question and shade in pencil on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only one answer to each question

1. What happens when ammonium chloride is heated?

A. dissolves.

B. melts.

C. sublimes.

D. cracks.

2. By which method is crude oil separated into its components?

A. fractional distillation.

B. filtration.

C. evaporation.

D. boiling.

3. Which is the smallest particle of an element that still retains its chemical properties?

A. Atom

B. Molecules

C. Compound

D. Ion

4. Which organelle is responsible for carrying out photosynthesis in plant cells?

A. Mitochondria

B. Nucleus

C. Chloroplast

D. Golgi apparatus.

5. How many electrons can be accommodated in the first energy level (n = 1)?

A. 2 electrons

B. 8 electrons

C. 18 electrons

D. 32 electrons

6. Which of these activities can disrupt the carbon cycle?

A. afforestation.

B. using solar energy.

C. using more petrol and diesel.

D. using nuclear energy.

7. Which life processes in plants work together to maintain the carbon balance in the air?

I. feeding II. reproduction III. respiration IV. photosynthesis

A. I and II only.

B. II and III only.

C. III and IV only.

D. I and III only

8. During which process is carbon removed from the atmosphere?

A. photosynthesis.

B. respiration.

C. decomposition.

D. combustion

9. For what purpose do female mosquitoes use blood during their life cycle?

A. They drink blood for nutrition.

B. They use blood to attract mates.

C. They use blood to lay eggs.

D. They use blood to develop wings.

10. What is the main source of carbon dioxide released into the atmosphere in the carbon cycle?

A. Volcanic eruptions.

B. Forest fires.

C. Respiration by plants and animals.

D. Photosynthesis

11. Which type of cell has a more complex internal organization?

A. Prokaryotic cells.

B. Eukaryotic cells.

C. Both prokaryotic and eukaryotic.

D. It varies depending on the organism.

12. An atom has 11 protons and 12 neutrons. How many electrons will it have in its neutral state?

A. 12

B. 11

C. 23

D. 0

13. Which gas contributes to global warming?

A. nitrogen dioxide.

B. carbon dioxide.

C. oxygen.

D. nitrogen.

14. Bengo claps and kills mosquitoes between his palms. Which method of mosquito control is he using?

A. chemically.

B. biologically.

C. physically.

D. genetically.

15. Which of the following is not a characteristic of a mixture?

A. It can be separated by physical means.

B. It has a fixed melting point.

C. Its properties are a combination of its components.

D. Its composition can vary.

16. Which of the following organisms is a unicellular eukaryote?

A. Algae

B. Amoeba

C. Archaea

D. Fungi

17. How many types of teeth are typically found in the mammalian tooth structure?

A. 1

B. 2

C. 3

D. 4

18. Which of the following is an example of an organic rock?

A. Granite

B. Limestone

C. Marble

D. Coal

19. Which type of rock is formed when existing rocks are transformed?

A. Sedimentary rock

B. Metamorphic rock

C. Igneous rock

D. Organic rock

20. Which of these is a sustainable practice that helps retain water in the soil and reduce water wastage in agriculture?

A. Using large amount of chemical fertilizers.

B. Employing excessive irrigation methods.

C. Mulching to cover the soil and reduce evaporation.

D. Using genetically modified seeds.

21. Which soil type is most likely to suffer from waterlogging and lead to poor plant growth?

A. Sandy soil.

B. Clay soil.

C. Loamy soil.

D. Rocky soil.

22. Which of the following is a potential solution to mitigate climate change?

A. Increasing the use of fossil fuels for energy production.

B. Planting more trees to absorb carbon dioxide.

C. Burning more coal and oil to create heat.

D. Disregarding renewable energy sources.

23. What is the process of using scientific knowledge to create something new or to improve existing products and processes called?

A. Experimentation.

B. Invention.

C. Observation.

D. Hypothesis

24. How does wearing face masks help to prevent the transmission of communicable diseases like COVID-19?

A. By reducing oxygen intake.

B. By creating a barrier that blocks the entry of pathogens.

C. By keeping the mouth and nose warm.

D. By improving lung function.

25. Which of the following is not a symptom of communicable diseases?

A. Fever.

B. Cough

C. Headache.

D. Happiness

26. Which of the following is a sustainable way of disposing of hazardous waste?

A. Dumping it in water bodies.

B. Burning it in open areas.

C. Recycling it for other uses.

D. Burying it in landfills.

27. Which of the following is an example of a renewable resource often used in sustainable waste management?

A. Fossil fuel.

B. Plastic bags.

C. Glass bottles.
D. Solar energy.

28. Which type of lever is the most common in the human body?
A. First-class lever.
B. Second class lever
C. Third class lever.
D. Fourth class lever.

29. Which simple machine is made of a grooved wheel with a rope or chain running along the groove?
A. Lever.
B. Wheel and axle
C. Inclined plane
D. Pulley.

30. Which device is used to increase or decrease the voltage of an alternating current?
A. Transformer.
B. Generator.
C. Motor
D. Solenoid

31. What is the process of producing an electric current in a wire by moving it through a magnetic field called?
A. Induction
B. Conduction
C. Repulsion
D. Attraction

32. What is the space around a magnet where the strength and direction of the magnetic field can be felt called?
A. Magnetic pole.
B. Magnetic field
C. Magnetic field lines
D. Magnetic force.

33. What is the basic S.I. unit of electric power?
A. Volts
B. Watts
C. Amperes
D. Ohms.

34. Which source of energy is considered non-renewable because it is obtained from ancient fossilized plants and animals?
A. Coal
B. Wind energy
C. Biomass energy
D. Geothermal energy

35. Which type of farming is commonly practiced in developed countries where advanced agricultural techniques are used?
A. Mixed farming
B. Subsistence farming
C. Commercial farming
D. Extensive farming.

36. Which type of organism is always found at the base of a food chain or food web?
A. Herbivore
B. Carnivore
C. Producers.
D. Decomposer.

37. Which planet is referred to as the "ice giant" because of its icy composition?

A. Jupiter

B. Saturn

C. Uranus

D. Neptune

38. Which type of teeth are located at the back of the mouth and are used for grinding and crushing food?

A. Incisors.

B. Canine's

C. Premolars

D. Molars.

39. What is the main purpose of hardening off seedlings raised on seedbeds before transplanting them?

A. To protect seedlings from pests and diseases.

B. To provide additional nutrients to the seedlings

C. To acclimate the seedlings to outdoor conditions.

D. To increase the size of seedlings.

40. Which type of lever has the load located between the fulcrum and the effort force?

A. First class lever.

B. Second class lever

C. Third class lever

D. Fourth class lever.

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BECE 2027 STEPPING STONE MOCK

SCIENCE

FINAL MARKING SCHEME

SECTION A (OBJECTIVE TEST) [40MARKS]

1. C	6. C	11. B	16. B	21. B	26. D	31. A	36. C
2. A	7. C	12. B	17. D	22. B	27. D	32. B	37. C
3. A	8. A	13. B	18. D	23. B	28. C	33. B	38. D
4. C	9. C	14. C	19. B	24. B	29. D	34. A	39. C
5. A	10. C	15. B	20. C	25. D	30. A	35. C	40. B

1. a) **(i) Name the parts labelled I, II, III, IV, V and VI.**
- I: Evaporating basin / Evaporating dish / Porcelain basin
- II: Mixture / Salt solution / Salt and water
- III: Tripod stand
- IV: Wire gauze / Wire mesh
- V: Bunsen burner / Spirit lamp / Source of heat
- VI: Steam / Water vapour / Escaping solvent / Water vapour

[½ mark each = 3 marks]

(ii) What method of separation is Kofi using?
- Evaporation / Evaporation to dryness [1 mark]

(iii) What will Kofi obtain in part I after heating the mixture in II for some time?
- Salt / Solid salt / Salt residue / Crystals of salt [1 mark]

(iv) State the function of the parts labelled:

(α) IV
- Distributes heat evenly / Spreads heat / Prevents the basin from cracking / Supports the basin

(β) V
- Provides heat / Heats the mixture / Supplies heat energy for evaporation

[1 mark each = 2 marks]

(v) **Why can the component obtained in I not be obtained by filtration of the mixture in II?**
Salt is soluble in water . Particles are too small and will pass through filter paper. [1 mark]

(vi**) What change would Kofi need to make to the apparatus if he also wanted to collect the component that escapes at VI? What would the new method be called?**

Change: Cover the basin with a lid / Connect a delivery tube from the basin to a condenser / Use a distillation apparatus / Set up a cooled receiver / Use Liebig condenser to condense the vapour / Place ice on top of lid to condense vapour.

New method: Distillation / Simple distillation / Distillation with condensation.
[1 mark for change = 1 mark, 1 mark for new method = 1 mark] = 2 marks

[TOTAL = 10 marks]

b) **(i) Name the parts labelled I, II, III, IV and V.**
- I: Cell wall
- II: Cell membrane / Plasma membrane (Accept: Cytoplasm if II points to the space)
- III: Chloroplast
- IV: Nucleus
- V: Vacuole / Central vacuole / Large vacuole / Cell sap vacuole

[1 mark each = 5 marks]

(ii) Give one function each of the parts labelled III, IV and V.
III (Chloroplast): Site for photosynthesis / Contains chlorophyll to trap sunlight for photosynthesis / Manufactures food for plant.
IV (Nucleus): Controls all activities of the cell / Controls cell division / Contains genetic material / Stores hereditary information.
V (Vacuole): Stores water and dissolved substances / Stores cell sap / Maintains turgidity of cell / Stores food, salts, waste products.

[1 mark each = 3 marks]

(iii) State:
(α) one part that is present in the plant cell but absent in an animal cell:
Cell wall / Chloroplast / Large central vacuole / Chlorophyll [1 mark]

(β) one difference between the vacuole in a plant cell and that in an animal cell:

- Plant cell has one large central vacuole / single prominent vacuole whereas animal cell has many small / tiny / scattered / temporary vacuoles.
- Plant cell vacuole is permanent while animal cell vacuole is temporary / small.
- Vacuole in plant cell is larger / more prominent than in animal cell.

Any one point - [1 mark]

c) **(i) Name the crops/plots labelled I, II, III and IV.**
- I: Legume / Groundnut / Beans
- II: Maize / Cereal / Corn
- III: Cassava / Yam / Root crop
- IV: Fallow

[0.5 mark each = 2 marks]

(ii) To which group of crops does the crop labelled
(α) I belong?
- Leguminous crops / Legumes / Nitrogen-fixing crops
(β) II belong?
- Cereal crops / Cereals / Grain crops / Grass family
(γ) III belong?
- Root and tuber crops / Root crops / Tubers

[1 mark each = 3 marks]

(iii) A farmer decides to plant the crop labelled II on the same plot for 4 consecutive years without rotating. Predict two problems he is likely to face.
- Depletion of specific soil nutrients / Loss of soil fertility / Nutrient imbalance
- Build-up of pests and diseases / Increase in pests/diseases that attack cereal
- Build-up of weeds / Weeds that attack maize will increase
- Low yield / Poor growth / Reduction in yield
- Soil erosion / Soil structure deterioration

Any two points - [1 mark each = 2 marks]

(iv) Which of the labelled crops helps to improve soil fertility naturally? Explain how it does it.
Crop: Crop I / Leguminous crop / Beans / Groundnut
Explanation: Fixes atmospheric nitrogen into the soil / Has root nodules containing Rhizobium bacteria that fix nitrogen / Adds nitrogen to the soil through symbiotic nitrogen fixation / Decays to add organic matter.

[1 mark for crop, 1 mark for explanation = 2 marks]

(v) Why is the plot labelled IV important in this rotation system even though no crop is harvested from it?
- Allows land to rest and regain fertility / Restores lost nutrients / Replenishes organic matter
- Prevents soil erosion / Covers soil
- Helps to break pest and disease cycle / Controls weeds, pests and diseases
- Improves soil structure / Conserves soil moisture

Any one point - [1 mark]

d)

i) **Identify the diagram above. [1mark]**
A pendulum

ii) **Identify the conversion of energy involved in the diagram, from A to D. [3marks]**
Potential energy → kinetic energy → Potential energy

iii) **Which type of energy is greatest at position B [1mark]**
Kinetic energy

iv) **Where in the diagram is potential energy equal to zero? [1mark]**
Point B

v) **What type of energy is the object having at point D? [1mark]**
Potential energy

vi) **If the mass of the object is 2.5kg and its raised 10m above the ground at point D, what is the magnitude of the energy possessed by the object? [Take g = 10ms^2] [3marks]**
The object possess Potential energy at point D = mass × acceleration due to gravity × height M1
= 2.5 × 10 × 10 M1
= 250J A1

2**. (a) Give two examples of each of the following types of substances:**
(i) element:
- Iron
- Oxygen
- Hydrogen
- Sodium
- Copper

- Carbon
- Aluminium

[1 mark each for any 2 = 2 marks]

(ii) compound:

- Water
- Carbon dioxide
- Sodium chloride
- Calcium carbonate
- Ammonia
- Glucose

[1 mark each for any 2 = 2 marks]

(iii) mixture:

- Air
- Salt solution
- Sugar solution
- Soil
- Sea water
- Brass
- Blood

[1 mark each for any 2 = 2 marks]

(b) i. Explain the following:

(α) renewable energy source [1 mark]

A renewable energy source is a source of energy that is naturally replenished on a human timescale and cannot be exhausted. It is obtained from natural processes that are continuously available and can be used over and over again without getting finished.

(β) non-renewable energy source [1 mark]

A non-renewable energy source is a source of energy that exists in a finite quantity in nature and cannot be replaced or replenished within a short period of time once it is used up. When it is exhausted, it is gone for millions of years.

ii. List two examples each of renewable energy source and non-renewable energy source:

Renewable energy source:

- Solar energy
- Wind energy
- Hydro energy
- Biomass energy
- Geothermal energy
- Tidal energy

[½ mark each for any 2 = 1 mark]

Non-renewable energy source:

- Coal
- Crude oil
- Natural gas
- Nuclear energy
- Petroleum

[½ mark each for any 2 = 1 mark]

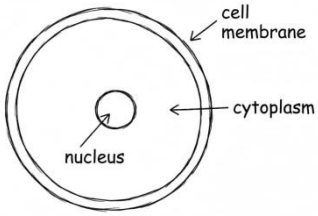
(c) **(i) What is a cell? [1 mark]**

A cell is the basic, smallest and fundamental unit of structure and function of all living organisms. It is the building block of life and all living things are made up of one or more cells.

(ii) Draw and label a typical animal cell to show: nucleus, cell membrane and cytoplasm

- Correct diagram - roughly circular
- Nucleus - correctly shown and labelled
- Cell membrane - correctly shown and labelled

- Cytoplasm - correctly shown and labelled



[Correct drawing = 1 mark, Any 2 correct labels = 1 mark = 2 marks]

(iii) State one function of the nucleus.

- Controls all activities of the cell
- Controls cell division and growth
- Stores genetic material / DNA
- Carries hereditary information from parents to offspring

[1 mark each for any 2 = 2 marks]

(d) i) Explain the following terms associated with animal feeding:

(α) Ration [1.5 marks]

Ration is the total quantity of feed that is given to a farm animal within a 24-hour period or in a day. It is carefully measured to provide all the nutrients the animal needs in the right proportion for maintenance, growth, and production.

(β) Supplement [1.5 marks]

A supplement is an additional feed or nutrient that is added to the main or basal feed of an animal to make up for nutrients that are lacking or deficient in the main feed. It is used to improve the nutritive value and balance the diet of the animal.

ii) List four most common ways of feeding farm animals.

- Stall feeding / Zero grazing
- Free range system / Extensive system
- Semi-intensive system
- Tethering
- Paddock grazing
- Rotational grazing
- Pasture grazing

[½ mark each for any 4 = 2 marks]

3. **(a) Name two:**

(i) deficiency diseases in humans:

- Kwashiorkor
- Marasmus
- Rickets
- Scurvy
- Anaemia
- Beriberi
- Night blindness
- Pellagra
- Goitre

[1 mark each for any 2 = 2 marks]

(ii) diseases caused by fungi in humans:

- Ringworm / Tinea
- Athlete's foot
- Candidiasis / Thrush
- Dhobi itch / Jock itch

[1 mark each for any 2 = 2 marks]

(b) (i) What is a binary compound? [2 marks]

A binary compound is a chemical compound that is composed of only two different elements. It is formed when two different elements combine together chemically in a fixed proportion. For example, sodium chloride is made of sodium and chlorine only.

(ii) Write the chemical formula of each of the following compounds:

(α) Sodium chloride - NaCl
(β) Ammonia - NH_3
(δ) Hydrochloric acid - HCl
(γ) Carbon dioxide - CO_2

[1 mark each = 4 marks]

(c) (i) What is a balanced ration in poultry feed? [2 marks]

A balanced ration in poultry feed is a feed that contains all the essential nutrients required by poultry birds in the correct quantity and in the right proportion to meet their needs for maintenance, growth, egg production and health. It contains carbohydrates, proteins, fats, vitamins, minerals and water in the correct amounts.

(ii) Give three effects of malnutrition in poultry birds.

- Stunted growth / poor growth / reduced weight gain
- Low egg production / drop in egg laying
- Poor feather development / feather loss
- Weak bones / leg deformities / rickets
- High mortality / increased death rate
- Increased susceptibility to diseases / low resistance to diseases
- Production of small-sized eggs / thin-shelled eggs
- Cannibalism and pecking of each other

[1 mark each for any 3 = 3 marks]

(d) i) Name the planets that constitute the inner planets of the solar system.

- Mercury
- Venus
- Earth
- Mars

[½ mark each for any 4 = 2 marks]

ii) Mention three properties of the inner planets.

- They are also called terrestrial planets / rocky planets
- They are small in size / have small diameter
- They have solid, rocky surfaces
- They are closest to the sun
- They have few or no moons - Mercury and Venus have no moons
- They have high density
- They have no ring system / they do not have rings
- They are made up of metals and silicate rocks
- They have thin or no atmosphere compared to outer planets

[1 mark each for any 3 = 3 marks]

4. **(a) Name two:**

(i) micro-nutrients of plants:

- Iron (Fe)
- Zinc (Zn)
- Boron (B)
- Manganese (Mn)
- Copper (Cu)
- Molybdenum (Mo)
- Chlorine (Cl)
- Nickel (Ni)

[1 mark each for any 2 = 2 marks]

(ii) macro-nutrients of plants:

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)

- Calcium (Ca)
- Magnesium (Mg)
- Sulphur (S)
- Carbon (C)
- Hydrogen (H)
- Oxygen (O)

[1 mark each for any 2 = 2 marks]

(b) i) Briefly describe a self-sustaining ecosystem and give two examples. [3 marks]
Description [2 marks]:
A self-sustaining ecosystem is an ecosystem that can maintain itself and continue to function on its own without much external support or interference from man. It is able to produce its own food, recycle its nutrients and wastes, and maintain a balance between all its living and non-living components through natural processes.
Examples [1 mark]:
- A natural forest
- A lake / pond
- An ocean / sea
- A balanced aquarium
- A grassland / savanna

[Any 2 examples = 1 mark]

ii) Distinguish between biotic and abiotic component of an ecosystem [2 marks]

Biotic components are the living parts of an ecosystem. They include all living organisms such as plants, animals, bacteria and fungi that can grow, reproduce and die. On the other hand, **Abiotic components** are the non-living parts of an ecosystem. They include physical and chemical factors such as sunlight, water, air, soil, temperature and minerals that support life but are not alive.

[1 mark for biotic, 1 mark for abiotic = 2 marks]

(c) (i) Distinguish between a colloid and a suspension. [2 marks]
A colloid is a mixture in which tiny particles of one substance are evenly dispersed and suspended throughout another substance but do not settle on standing. The particles are small enough that they cannot be seen with the naked eye and cannot be separated by ordinary filter paper. Example: milk. On the other hand, **a suspension** is a heterogeneous mixture in which larger, insoluble solid particles are dispersed in a liquid or gas but will eventually settle down on standing if left undisturbed. The particles are large enough to be seen and can be separated by filtration. Example: sand and water.

[Correct distinction = 2 marks]

(ii) Classify the following as solution, colloid or suspension:
(α) Salt and water - **Solution**
(β) Milk and water - **Colloid**
(γ) Sand and water - **Suspension**

[1 mark each = 3 marks]

(d) State two effects of poor waste management on human health. [2 marks]
- Causes outbreak of diseases such as cholera, typhoid, dysentery, malaria
- Causes air pollution leading to respiratory diseases
- Contaminates water sources leading to water-borne diseases
- Breeding of disease vectors such as mosquitoes, flies, rats, cockroaches
- Causes bad odour / unpleasant smell
- Can cause injury from sharp objects / cuts
- Causes contamination of food

[1 mark each for any 2 = 2 marks]

(e) (i) What is work input? [2 marks]

Work input is the total amount of energy or effort that is put into a machine to make it work or operate. It is the work done on the machine by the effort applied.
Formula: Work Input = Effort x Effort Distance

(ii) Explain how work output affects the efficiency of a machine if work input is fixed. [2 marks]
Efficiency of a machine is calculated as: Efficiency = (Work Output / Work Input) x 100%. If work input is fixed, it means the denominator is constant. Therefore, when work output increases, the efficiency of the machine also increases. When work output decreases, the efficiency decreases. So work output is directly proportional to efficiency when work input is fixed.

[Correct explanation = 2 marks]

5. a) (i) What is green economy? [2marks]

Green economy is defined as an economy that aims at reducing environmental risks and ecological scarcities and that aims for sustainable development without degrading the environment.

[2marks]

(ii) State three climate change adaptation measures that you can apply to conditions in your community. [3marks]

- Using scarce water resources efficiently
- Planting drought – tolerant crops
- Building flood defences and raising levels of dykes
- Selecting tree species that can absorb high Carbon (IV) oxide from the atmosphere and release enough oxygen.
- Building codes for future climate condition and extreme weather events

[1mark each for any 3]

b) (i) Name the main stages involved in electricity power supply. [3marks]

- Generation stage
- Transmission stage
- Distribution stage

[1mark each]

(ii) What is the function of a capacitor in an electronic circuit? [2marks]

A capacitor stores electrical energy in a circuit.

c) (i) **State three non- metals in the first twenty elements.**

- Hydrogen
- Carbon
- Nitrogen
- Oxygen
- Fluorine
- Phosphorus
- Sulphur
- Chlorine

[3 marks @ 1 mark each]

(ii) **Give three properties of non-metal.**

- Non- metals are brittle
- They are bad conductors of heat and electricity.
- They cannot be polished
- They are neither tough nor strong.

[3 marks @1 marks]

d) (i) **Mention two breeding habit of a house fly.**

- Garbage's
- Decaying organic materials.
- Rotten fruits.
- Vegetables
- Solid organic waste in open drains.

[2 marks @ 1 mark each]

(ii) **State two possible diseases that housefly can cause.**

- Typhoid fever
- Dysentery
- Cholera
- Tuberculosis
- Yaws
- Poliomyelitis
- Conjuretivities
- Salmonellosis.

- Anthrax
- Tularemia

[2 marks @ 1 mark each]
Correct spellings for score